

Road Accident Simulation Report

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Accident Type: Rear-end collision
Location: Bulhao, Labo, Camarines Norte
Time of Accident: 10.01 seconds into simulation
Cause: Poor Visibility
Weather: Cloudy
Road Geometry: straight (Slope: 0.0°)

Vehicles Involved:

- Car 1: Mass = 1206 kg, Initial Speed = 57.0 km/h
- Car 2: Mass = 1367 kg, Initial Speed = 54.9 km/h

Analysis:

This simulation demonstrates how speed, road conditions, and lighting contribute to accidents. Severity is classified based on impact force: minor (<50kN), moderate (50-150kN), severe (>150kN).

Recommendations:

- Install anti-skid road surfaces
- Upgrade street lighting
- Improve weather-responsive lighting systems
- Maintain current lighting standards
- Increase following distance education

Systems Interaction Summary:

- Poor visibility limits human perception, interacting with vehicle lighting and road markings in systems theory to compromise safety margins.

Intervention Scenario Comparison:

- Baseline Severity Distribution: severe: 17, moderate: 2, none: 1
- Intervention Severity Distribution: severe: 14, moderate: 4, none: 2
- Baseline Avg Impact Force: 1284229.83 N
- Intervention Avg Impact Force: 982779.81 N
- Avg Risk Score Reduction: 6.2% (9.60 -> 9.00)
- Avg Impact Force Reduction: 23.5%
- Interventions Applied: Road Surface Treatment

Machine Learning Prediction:

ML Prediction disabled